HEET NITINKUMAR DALSANIA

Phoenix, Arizona | heet.dalsania@outlook.com | github.com/heetdalsania | heetdalsania.github.io | LinkedIn

RESEARCH PROFILE

Computer science student and undergraduate researcher working across vision-language models, AI evaluation, multimodal perception, and reliable software systems. My work combines careful experimental design with practical tools and infrastructure for making AI useful and accountable.

EDUCATION

Arizona State University, Barrett, The Honors College — B.S. Computer Science, Ira A. Fulton Schools of Engineering; expected May 2028 (three-year plan). GPA: 3.98/4.00. Dean's List. Tempe, AZ.

RESEARCH EXPERIENCE

People and Robots Laboratory (PeRL), Arizona State University — Undergraduate Researcher | Jan 2026–present

- Advisor: Prof. Pooyan Fazli. Designed and ran a controlled vision-language-model evaluation over 1,000 videos to test whether explanations of AI-generated video are grounded in visual evidence.
- Audited the analysis pipeline, quarantined invalid intermediate results, and rebuilt joins around video identifiers with manifest checks and run-level provenance.
- Contributed benchmark validation to 360QVR, a dataset for spatial and temporal reasoning in 360-degree video; publication outcome not yet confirmed.

Barrett Undergraduate Research Fellowship, Arizona State University — Research Fellow | May–Dec 2025

- Advisor: Prof. Jianming Liang. Continued biomedical-imaging and computer-aided-diagnosis research.

ASU SCENE Program — Undergraduate Researcher | Sep 2024–May 2025

- Advisor: Prof. Jianming Liang. Developed and evaluated label-efficient CLIP adaptation for chest X-ray disease classification; led method design, benchmarking, and sole-author write-up.

RESEARCH OUTPUTS

Forensic Reasoning or Fluent Confabulation? Evaluating Vision-Language Model Grounding in AI-Generated Video Detection — sole-author manuscript in preparation

- Controlled comparisons reveal that a vision-language model can produce confident, specific but conflicting explanations for identical video frames when supplied an incorrect label. The analysis includes paired tests and cross-model evaluation; no publication is claimed.

360QVR: Quantitative Angular Reasoning for Vision-Language Models in 360° Videos — co-author; submitted to NeurIPS 2026 Evaluations & Datasets Track

- Contributed quality control for segmentation, spatial grounding, and temporal continuity. Current decision status unverified.

Label-Efficient Chest X-ray Diagnosis via Partial CLIP Adaptation — sole-author arXiv preprint 2507.07254 (2025; not peer reviewed)

- Partially adapted CLIP ViT-B/32 using approximately 25% of its visual-encoder parameters. Achieved mean AUC 0.750 across 14 NIH ChestX-ray14 disease classes without paired report supervision.

SELECTED SOFTWARE AND AI PROJECTS

Quire — local-first collaborative Markdown workspace | TypeScript, Yjs, Node.js, MCP

- Keeps Markdown files on disk as the source of truth while people and AI agents edit together with live cursors, authorship, reviewable suggestions, and enforced agent permissions. Published as the public quiredocs beta.
- Built a 17-tool MCP interface for agent participation and cross-platform continuous integration that tests the packaged release on Linux, macOS, and Windows.

FootballVision — football-video analysis | Python, PyTorch, OpenCV, FastAPI

- Detects and tracks players, maps positions into pitch coordinates with camera calibration, and produces tactical views. A benchmarked caching and calibration sweep reduced per-frame latency from 498 ms to 107 ms in the measured configuration.
- Uses object detection, ByteTrack identity tracking, pitch-keypoint homography, and structured spatial features to derive formation and pressing information from broadcast footage.

Physiq — nutrition and training application | React, JavaScript, Capacitor

- Supports food search, nutrient tracking, barcode scanning, workouts, and a 12-week projection. Uses public nutrition APIs directly, without an application backend; iOS packaging is in preparation.
- Built a bundling pipeline and hand-rolled SVG charts to avoid a browser-side compiler and unnecessary dependencies.

FADE — autonomous GitHub digest and cost analysis | Python, LLMs, React

- Generates pull-request digests and analyzes its own trace to separate deterministic, rule-based, and model-dependent operations so unnecessary model calls can be replaced with code.

Pulse — market-news intelligence | JavaScript, REST APIs, NLP

- Aggregates financial news for watchlists and ranks articles by relevance and lexicon-based bullish, bearish, or neutral sentiment.

NBA MVP Predictor — sports-data modeling | Python, pandas, scikit-learn

- Analyzes player statistics from 1995–2024 to model MVP-share outcomes and inspect the contribution of individual features.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL.

AI and evaluation: PyTorch, scikit-learn, CLIP, vision-language models, computer vision, transfer learning, few-shot learning, controlled experiments, statistical testing, LLM-as-judge evaluation.

Systems and data: Linux, Slurm, Git/GitHub Actions, pandas, DuckDB, PostgreSQL, REST APIs, AWS, Google Cloud.

Web and agents: FastAPI, Node.js, React/Next.js, Yjs CRDTs, Model Context Protocol, local model inference.

CERTIFICATIONS AND HONORS

Google Cloud Professional Machine Learning Engineer; AWS Certified AI Practitioner; Oracle Cloud Infrastructure Generative AI Professional; IBM Applied AI Professional Certificate; Stanford/DeepLearning.AI Supervised Machine Learning; Barrett Undergraduate Research Fellow; ASU Dean's List; AP Scholar; National Latin Exam (two Gold, one Silver).